

Java suggestion cse-2

Define Object-Oriented Programming. Explain its main features with examples.

Explain the concept of encapsulation. How is it implemented in Java?

What is abstraction? Differentiate between abstraction and encapsulation.

Explain inheritance in Java. Discuss its types with examples.

What is polymorphism? Explain compile-time and runtime polymorphism.

Differentiate between method overloading and method overriding.

Explain dynamic method dispatch in Java with an example.

Explain the concept of constructor. Differentiate between default and parameterized constructors.

What is constructor overloading? Explain with example.

What is access control in Java? Explain different access specifiers.

Explain the use of this keyword in Java.

What is super keyword? Explain its uses with examples.

What is final keyword? Explain its use with variables, methods, and classes.

Explain the process of compiling and executing a Java program.

Explain arrays in Java. How are they declared and initialized?

What is String class in Java? Explain its features and methods.

What is garbage collection in Java? How does it work?

What is an object? How is it created in Java? Explain with syntax and example

What are instance variables, local variables and class variables?
Explain with suitable example.

What is a constructor? Explain its role in object initialization.

What is constructor overloading? Illustrate with a program.

Explain the use of this keyword in Java with examples.

Explain nested classes (inner classes) in Java

What is method overloading within a class? Explain with example.

Explain different types of inheritance supported in Java with examples.

Explain multilevel inheritance with an example.

What is hierarchical inheritance? Give an example.

Why does Java not support multiple inheritance using classes?

What is method overriding? Explain with an example.

What is the difference between method overloading and method overriding?

Differentiate between compile-time and runtime polymorphism.

Explain dynamic method dispatch with a suitable example.

What is a package in Java? Explain its importance.

What are the steps to compile and run a package program?

Explain built-in packages in Java with examples.

What is the difference between user-defined and predefined packages?

Explain access protection in packages (public, private, protected, default).

Differentiate between checked and unchecked exceptions with examples.

Explain the use of try, catch, and finally blocks with syntax.

What is the purpose of the finally block? When is it executed?

Explain the use of the throw keyword with an example.

What is the throws keyword? How is it different from throw?

Explain multiple catch blocks with an example.

What is the difference between Error and Exception?

What is ArithmeticException, NullPointerException, and ArrayIndexOutOfBoundsException?

Explain the life cycle of a thread in Java.

What are the different ways to create a thread in Java?

Differentiate between Thread class and Runnable interface.

Explain the use of start() and run() methods.

What is an ArrayList in Java? Explain its features.

Explain the difference between Array and ArrayList.

What is the difference between ArrayList and Vector?

What is a Vector in Java? Explain its features.

How is Vector different from ArrayList?

Explain the role of Frame and JFrame in Java GUI.

Explain the concept of layout managers in Java.

Describe different types of layout managers (FlowLayout, BorderLayout, GridLayout).